

Notes: Harford (JFE, 2005)

What Drives Merger Waves?

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1 Big picture

- **Research question.** Why do merger waves happen? Are aggregate merger waves mainly caused by market timing and stock-market misvaluation, or by real economic, regulatory, and technological shocks that require industries to reorganize assets?
- **Main idea.** Industry merger waves need two ingredients: a real shock that creates an economic reason to reallocate assets, and enough capital liquidity to finance the transactions. Industry shocks create the motivation; capital liquidity allows the wave to happen.
- **Core contrast.** The behavioral view says merger waves arise because overvalued bidders use stock to buy less-overvalued firms. The neoclassical view says waves arise because industries respond to shocks by reorganizing assets through mergers and acquisitions.
- **Harford's modification.** The neoclassical story needs a macro liquidity component. Shocks alone are not enough. When transaction costs and financing constraints are high, even real shocks may not generate a wave.
- **Main finding.** Economic shocks, deregulation, and capital liquidity predict industry merger waves. Market-timing variables have little additional explanatory power once these neoclassical variables are included.
- **Aggregate implication.** Aggregate merger waves occur because many industry-level waves cluster when macro capital liquidity is high. The clustering of aggregate activity does not require shocks themselves to cluster strongly in time.
- **Key evidence against pure market timing.** Cash-financed partial-firm acquisitions rise during merger waves, and even stock bidders are more likely to make cash purchases of divisions. This is hard to reconcile with a story based only on using overvalued equity.
- **Return evidence.** Industry returns do not show robust abnormal underperformance after waves once Fama–French factors and equal weighting are used. Bidder underperformance appears only weakly and mainly in value-weighted stock-bidder specifications.
- **Operating-performance evidence.** Mergers in waves do not perform worse than mergers outside waves; some measures, including sales growth and analyst long-term growth forecasts, look better in waves.
- **Economic takeaway.** Merger waves are mostly real reallocation episodes. High valuations are correlated with waves not necessarily because of mispricing, but because booms and high asset values are also periods of high capital liquidity.

2 Incremental contribution of the paper

- **Direct comparison of theories.** Earlier papers often test either behavioral market-timing explanations or neoclassical shock explanations. Harford explicitly puts both in the same empirical framework.
- **Industry-level wave identification.** The paper identifies merger waves at the 48-industry level rather than studying only aggregate merger counts. This makes it possible to ask whether waves are linked to industry-specific shocks.

- **Capital liquidity as the missing link.** The paper adds a liquidity condition to the neoclassical model. Shocks create demand for reallocation, but financing conditions determine whether the reallocation happens through a concentrated wave of mergers.
- **Uses partial-firm acquisitions as a sharp test.** Divisional acquisitions for cash should rise during neoclassical reallocation waves, but they are not naturally predicted by a pure overvalued-stock story.
- **Connects industry waves to aggregate waves.** The paper shows that aggregate merger waves are driven by the clustering of identified industry waves, not simply by economy-wide market timing.
- **Separates liquidity from valuation.** The paper argues that the observed link between high market valuations and merger waves is largely a liquidity effect rather than direct evidence of misvaluation.
- **Return and performance tests.** The paper supplements wave-prediction tests with post-wave return tests, bidder return tests, operating performance tests, and analyst forecast revisions.

3 Data and measurement

3.1 Merger sample

The paper starts with merger or tender-offer bids from Thomson Financial’s Securities Data Company (SDC).

- **Period.** 1981–2000.
- **Transaction cutoff.** Deals must have transaction value of at least \$50 million.
- **Industry assignment.** Each bidder and target is assigned to one of 48 Fama–French industries using the SIC code recorded by SDC at announcement.
- **Counting rule.** If the bidder and target are in different industries, the deal counts toward merger activity in both industries. If both are in the same industry, it counts once for that industry.
- **Decade split.** The sample is split into the 1980s and 1990s because aggregate merger activity has distinct waves in those two periods, with a trough around the 1990–1991 recession.

Interpretation. The paper is not studying isolated merger announcements. It is studying clustering in merger activity within industries over time.

3.2 Identifying industry merger waves

For each industry and decade, Harford identifies the 24-month period with the highest concentration of merger activity. The wave test compares that actual concentration with simulated concentrations.

- For each industry-decade, the actual number of bids is randomly distributed across 120 months 1,000 times.

- For each simulated sample, the maximum 24-month concentration is recorded.
- The actual 24-month cluster is classified as a wave if it exceeds the 95th percentile of the simulated distribution.
- The final sample contains 35 waves from 28 industries.
- Seven industries have two waves, one in the 1980s and one in the 1990s.
- The average number of bids in a 24-month non-wave period for the wave industries is 7.8, while the average number during a 24-month wave is 34.3.

Interpretation. The wave measure asks whether activity is unusually clustered relative to what one would expect from the industry's own total merger activity in that decade.

3.3 Industry shock variables

The neoclassical hypothesis predicts that merger waves should be preceded by real shocks to the industry's operating environment. The paper measures economic shocks using changes in:

- cash-flow margin on sales;
- asset turnover;
- research and development scaled by beginning-of-period assets;
- capital expenditures scaled by beginning-of-period assets;
- employee growth;
- return on assets;
- sales growth.

Because different shocks may have different signs across industries, the paper uses the median absolute change in each variable. The pre-wave value is then ranked relative to the same industry's own 20-year history.

Interpretation. The test is not whether sales always rise or always fall before waves. It is whether industries experience unusually large changes before waves.

3.4 Behavioral and valuation variables

The behavioral hypothesis predicts waves after periods of high valuation and high valuation dispersion. The paper therefore studies:

- industry market-to-book ratios;
- changes in market-to-book ratios;
- intra-industry dispersion in market-to-book ratios;
- average one-year and three-year stock returns;
- intra-industry dispersion in those returns.

Market-to-book is ambiguous because it can capture both growth opportunities and possible overvaluation. This is why the paper also looks at return dispersion and later adds market-to-book to logit models with real shock and liquidity variables.

3.5 Capital liquidity

The key macro liquidity proxy is the spread between the average commercial and industrial loan rate and the federal funds rate:

$$\text{C\&I Rate Spread}_t = \text{C\&I loan rate}_t - \text{Federal funds rate}_t.$$

- The spread comes from the Federal Reserve’s Survey of Terms of Business Lending.
- A high spread indicates tighter credit and lower capital liquidity.
- A low spread indicates easier financing and higher capital liquidity.
- The paper also uses this spread to distinguish shocks that occur when capital is tight from shocks that occur when capital is liquid.

Interpretation. This measure helps separate a liquidity explanation from a valuation explanation. High valuations can be correlated with merger waves because high valuations often coincide with low financing constraints.

3.6 Additional outcome variables

The paper also examines several outcomes that help distinguish the two theories:

- **Partial-firm acquisitions.** The paper uses divisional or partial-firm acquisitions, especially cash-financed ones, as evidence of real asset reallocation.
- **Industry returns.** Calendar-time industry portfolios are used to test whether wave industries have abnormal returns before, during, or after waves.
- **Bidder returns.** The paper separately studies bidders, especially stock bidders inside and outside waves.
- **Operating performance.** The paper studies profitability, asset turnover, ROA, sales growth, market-to-book, and analyst long-term growth forecasts following mergers.

4 Methodology and empirical specifications

4.1 Univariate shock evidence

The first empirical step compares industry characteristics in the year before a wave with the same industry’s own historical distribution.

- Economic-shock variables are unusually large before waves.
- Deregulatory events are listed separately and often line up with identified wave industries.

- Market-to-book and dispersion in market-to-book are also high before waves.
- The return variables are weaker: average industry returns are high before waves, but not strongly abnormal relative to the industry's own return history.

Interpretation. The univariate tests show that waves are associated with both real shocks and high valuations, so later tests must put these variables in the same model.

4.2 Logit model for industry wave starts

The main wave-prediction model is a logit model of the beginning of an industry merger wave:

$$Pr(WaveStart_{i,t} = 1) = F(\alpha + \beta X_{i,t-1} + \gamma L_{t-1} + \delta X_{i,t-1} \times TightCapital_{t-1}),$$

where $X_{i,t-1}$ includes industry shock and valuation variables, and L_{t-1} is capital liquidity.

The main explanatory variables include:

- industry market-to-book;
- three-year industry return and dispersion in returns;
- C&I rate spread;
- deregulation dummy;
- economic shock index, the first principal component of the seven economic shock variables;
- interaction of economic shock index with a tight-capital dummy.

Interpretation. The key test is whether the economic shock index predicts waves more strongly when capital is liquid and whether market-timing variables still matter after these controls.

4.3 Aggregate merger wave model

The paper then asks whether identified industry waves explain aggregate merger activity. Aggregate merger activity is the fraction of firms in the economy involved in merger activity in a year.

- The fraction of bids in industries currently undergoing waves is highly correlated with aggregate merger activity.
- The reported correlation is 0.85.
- Aggregate logit models use industry-weighted versions of the same variables from the industry-wave model.

Interpretation. Aggregate merger waves are not treated as separate from industry waves. They are the result of many industry waves occurring at the same time.

4.4 Partial-firm acquisition tests

The paper uses partial-firm acquisitions to distinguish real reallocation from overvalued-stock acquisition.

- Panel correlations compare industry merger activity and partial-firm acquisition activity.
- Firm-level logit models ask whether merger bidders are also more likely to buy partial-firm assets.
- Cash partial-firm acquisitions are especially important because they do not fit a story in which waves are mainly overvalued-stock swaps.

Interpretation. If waves reflect real asset reallocation, mergers and divisional acquisitions should move together. If waves mainly reflect stock overvaluation, cash divisional acquisitions should not be central.

4.5 Return tests

The paper uses Fama’s calendar-time portfolio method. It estimates abnormal returns for industry portfolios and bidder portfolios before, during, and after waves.

- For industries, Fama–French factor-adjusted returns are not significantly different around waves.
- Unadjusted value-weighted returns show the familiar high-before-and-during, low-after pattern, but this is not robust to equal weighting.
- For bidders, there is some value-weighted evidence that stock bidders in waves underperform, but the effect is not robust to equal weighting and is not statistically different from non-wave stock bidders.

Interpretation. The return evidence does not provide strong support for the behavioral prediction of systematic post-wave reversal.

4.6 Operating-performance tests

The operating-performance regression is:

$$PostPerformance_j = \alpha + \rho PrePerformance_j + \theta InWave_j + \varepsilon_j.$$

The paper estimates this specification using individual observations and group means. It also adds a contractionary-wave dummy and uses analyst long-term growth forecast revisions.

- The wave dummy is not negative for post-merger performance.
- Sales growth is higher for wave mergers in several specifications.
- Analyst forecast revisions are more positive for mergers inside waves.
- Contractionary-wave controls do not overturn the main conclusion.

Interpretation. Mergers in waves do not look like value-destroying transactions driven only by temporary misvaluation.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether industry merger waves are systematically preceded by real shocks and whether those shocks propagate into waves only when capital liquidity is high. The core empirical variation is industry-time variation in shock exposure combined with time-series variation in macro liquidity.

5.2 Why the liquidity interaction matters

The interaction between industry shocks and capital liquidity is central. If shocks alone caused waves, liquidity should not matter. If market timing alone caused waves, real shocks should not matter much after valuation controls. Harford finds that both shocks and liquidity matter, and that shocks are less likely to generate waves when capital is tight.

5.3 Why market-to-book is not decisive evidence of misvaluation

Market-to-book can be high because firms are overvalued, but it can also be high because growth opportunities and asset values are high. In Harford's framework, high asset values relax financing constraints and increase capital liquidity. Therefore, the observed correlation between valuation and merger waves is not enough to prove market timing.

5.4 Why cash divisional acquisitions are powerful evidence

The behavioral hypothesis predicts stock-financed acquisitions by overvalued bidders. It does not naturally predict a simultaneous rise in cash-financed divisional acquisitions. The fact that cash partial-firm acquisitions rise during waves supports the real reallocation interpretation.

5.5 Limits of the evidence

The paper does not prove that no mergers are motivated by market timing. Instead, it argues that market timing does not explain merger waves as an aggregate phenomenon. The evidence is strongest against market timing as the primary driver of wave clustering.

6 Tables and figures

6.1 Table 1: Predictions of the neoclassical and behavioral hypotheses

Read:

- Table 1 summarizes the predictions of the two theories and compares them with the paper's findings.
- The neoclassical view predicts industry shocks, capital liquidity, cash partial-firm acquisitions, and no necessary post-wave underperformance.

- The behavioral view predicts high valuation, high valuation dispersion, stock-financed acquisition activity, and poor post-wave returns.
- The findings line up more closely with the neoclassical view modified to include capital liquidity.

Economic message: This table is the conceptual map for the paper. It clarifies what evidence would support each theory before moving to the empirical tests.

Table 1
Predictions of the neoclassical and behavioral hypotheses for merger waves

	Neoclassical	Behavioral	Finding
Cause of industry wave	Regulatory or economic shock accompanied by capital liquidity	Overvaluation and dispersion of valuation within industry	Regulatory and economic shocks accompanied by capital liquidity
Cause of aggregate wave	Multiple simultaneous industry waves clustering because of macro liquidity factor	Overvaluation and dispersion in the aggregate	Multiple simultaneous industry waves clustering because of macro liquidity factor
Cash partial-firm acquisitions	Increase during wave; could be made by stock bidders	Do not increase during wave; are not made by stock bidders	Increase during waves and are made by stock bidders
Pre-wave returns and market-to-book ratios	High if capital liquidity is tied to asset valuation	High	High
Dispersion in pre-wave returns	No prediction	High	Normal
Post-wave returns	No prediction	Low	Normal
Measures of tight credit	Low if capital liquidity is important	No prediction	Low
Post-merger operating	Better than without a	Worse in waves	Similar/better in waves

Table 1: Predictions of the neoclassical and behavioral hypotheses for merger waves.

6.2 Table 2: Industries with merger waves

Read:

- Table 2 lists the industries identified as having merger waves and the starting dates of those waves.
- The table includes 35 waves across 28 industries.
- Many reasons are regulatory or technological: banking deregulation, telecommunications deregulation, cable deregulation, retail consolidation, healthcare bargaining power, and the Internet.
- The reasons show that merger waves are often tied to concrete industry events rather than generic stock market conditions.

Economic message: The wave list makes the neoclassical mechanism tangible. Different industries experience different shocks, but high liquidity can make many of these shocks generate waves around the same time.

Table 2
Industries with merger waves
The industries and starting dates of the merger waves comes from the procedure described in Section 3. The reasons for the wave come from Lexis-Nexis searches of news reports analyzing the merger activity at the time of the wave.

Industry	Date wave started and reason given
Aircraft	Jan, 1999 Big, older fleets require increased maintenance, repair and overhaul Increasingly outsourced from carriers, who want "one-stop shops"
Banking	Aug, 1985 Deregulation allows interstate banking, particularly in California Oct, 1996 Deregulation and Information Technology (IT)
Business Services	Oct, 1986 Partially IT-driven mergers as IT becomes important Sep, 1998 Fragmented, smaller players combine, share cost structures, offer more complete line of services to customers—industry grows as outsourcing takes off
Business Supplies	Jan, 1997 Paper and pulp industry consolidates from fragmented price takers to gain market power and avoid costly duplication of capital intensive production facilities
Candy & Soda	Apr, 1992 Snapple and other non-carbonated beverages make strides, leading to activity to beat or buy them
Chemicals	Mar, 1995 Large cash flows, over capacity in production, need to consolidate research
Communication	Nov, 1987 Deregulation: Break-up of AT&T in 1984 followed by entry into long distance, investment in fiber optic capacity, etc. July, 1997 Deregulation: Telecommunications Act in 1996, consolidation, technological changes
Computers	July, 1998 Internet
Consumer Goods	Aug, 1986 Mature market and the need to offer full line leads to consolidation
Electrical Equipment	June, 1986 Several companies seek growth through acquisition to compete better with industry leaders Westinghouse and General Electric
Electronic Equipment	Jan, 1999 OEM's growth leads to demand for electronic equipment manufacturers to shift from small regional players to larger global players capable of infrastructure, IT, etc. to grow with their customers

Table 2, part 1: Early portion of industries with identified merger waves.

Table 2 (continued)

Industry	Date wave started and reason given
Entertainment	Oct, 1987 Deregulation allows firms to own many stations Mar, 1998 Studios seek diversified production sources and strong libraries; Telecom act of 1996 relaxes media ownership limits
Food Products	Jan, 1999 Retail consolidation pushes distribution consolidation and/or sale of distributors to bigger retailers who want to buy rather than build distribution channels
Healthcare	May, 1996 Service providers consolidate to have bargaining power with HMOs
Insurance	Nov, 1998 Bigger is safer, leading to consolidation, especially in reinsurers
Machinery	May, 1996 Large manufacturers decreased number of suppliers they were willing to deal with in bid to improve efficiency. This forced consolidation in a number of capital goods industries—many smaller players were bought in "roll-up" deals
Measuring and Control Equip	Nov, 1998 Depression in semi-conductor industry (big customer)
Medical Equipment	Nov, 1998 Two motives: first, acquisitions in core areas to grow, then acquisitions outside core areas to offer broad products to increasingly consolidated customer base (hospitals)
Personal Services	Feb, 1996 Consolidation in legal and funeral services industries
Petroleum and Natural Gas	June, 1997 Increasing prices, record drilling, increasing costs lead drive to increase size to be more efficient
Pharmaceutical Products	Oct, 1998 Mid-sized companies merge to garner size necessary to fund increasingly large costs of development
Restaurants, Hotels, Motels	Mar, 1985 Saturation and similarity, trends toward take-out, competition from supermarket delis Dec, 1996 Operators such as Starwood have buying sprees. Others buy properties to gain sufficient bulk to compete in corporate account business market
Retail	Oct, 1986 Shift to specialty stores as aging department stores consolidated; value of land & buildings in revitalized urban centers Aug, 1996 Strong growth and impact of internet

Table 2, part 2: Continued list of industries and wave rationales.

Table 2 (continued)

Industry	Date wave started and reason given
Shipbuilding, Railroad Equip	Aug, 1998 Shrinking defense budgets finally forced the issue of overcapacity in the industry
Steel Works	Sep, 1997 Collapse in demand from Asia leads to falling prices forcing consolidation
Transportation	Aug, 1986 Mostly still working out issues following deregulation July, 1997 End of Interstate Commerce Commission, overcapacity in shipping, open-skies agreements, railroad consolidation started with a few big mergers and then forced responses to balance
Utilities	Nov, 1997 Deregulation in some markets plus elimination of a law prohibiting mergers between non-contiguous providers
Wholesale	June, 1996 Simultaneous consolidation in several wholesale sectors as growth slows and firms move to add breadth, take advantage of new IT ability, grow by acquisition

Table 2, part 3: Final industries and wave rationales.

6.3 Table 3: Economic shocks, stock valuation, and deregulation

Read Panel A:

- Economic-shock variables are abnormally large before waves.
- Profitability, asset turnover, R&D, capital expenditures, employee growth, ROA, and sales growth have high pre-wave ranks relative to industries' own histories.
- Market-to-book and industry dispersion in market-to-book are also high before waves.
- Return variables are less conclusive than market-to-book variables.

Read Panel B:

- The panel lists major deregulation events during the sample period.
- Deregulatory events line up with several industries that experience waves, including banking, communications, transportation, utilities, entertainment, and oil and gas.

Economic message: Industry waves are preceded by large real changes and by identifiable regulatory events. High valuation is present too, but not enough by itself to distinguish theories.

measure economic shocks to the industry: net income/sales (profitability), asset turnover, R&D, capital expenditures, employee growth, ROA, and sales growth. The median absolute change in each of above variables is computed for each industry-year. Market-to-book and dispersion in market-to-book are either economic variables or misvaluation proxies. Stock valuation is also addressed by the median prior one- and three-year compounded return for firms in the industry along with the intra-industry dispersion of that return. For all variables, the number presented in the table is the mean, across all industries, of this industry-specific median in the year immediately preceding the start of the merger wave (there are 35 industry-year observations for this pre-wave year). For each industry, the 20-year time series of shock observations is ranked into quartiles and the cross-industry mean rank of the shock in the pre-wave year is presented. A test is performed on the average difference between a rank of 2.5 (middle) and the ranking of the pre-wave year within its own industry time series. The p -value for the hypothesis that this difference is zero is presented in brackets.

Economic shocks (Variables related to the neoclassical hyp.)	Mean	Rank	Stock valuation (Variables related to the behavioral hyp.)	Mean	Rank
Net income/sales H_0 : Rank = 2.5	0.030	2.83 [0.100]	3yr-Return H_0 : Rank = 2.5	0.638	2.66 [0.372]
Asset turnover H_0 : Rank = 2.5	0.096	2.77 [0.076]	σ (3yr-Return) H_0 : Rank = 2.5	1.317	2.60 [0.372]
R&D H_0 : Rank = 2.5	0.004	2.86 [0.007]	1yr-Return H_0 : Rank = 2.5	0.132	2.66 [0.328]
Capital expenditures H_0 : Rank = 2.5	0.023	2.80 [0.109]	σ (1-yr Return) H_0 : Rank = 2.5	0.566	2.74 [0.152]
Employee growth H_0 : Rank = 2.5	0.128	2.97 [0.008]	Market-to-Book variables (related to both hypotheses)		
			Market to book H_0 : Rank = 2.5	1.563	3.14 [0.001]
ROA H_0 : Rank = 2.5	0.040	2.80 [0.066]	Industry σ (market to book) H_0 : Rank = 2.5	1.201	3.11 [0.001]
Sales growth H_0 : Rank = 2.5	0.128	3.03 [0.003]	Change in market to book H_0 : Rank = 2.5	0.022	2.86 [0.055]

Table 3, Panel A: Economic shocks and stock-valuation variables before waves.

Table 3 (Continued)

(Panel B) This table lists the major deregulatory initiatives during the sample period and is constructed from Viscusi et al. (2000), Economics of Regulation and Antitrust, Tables 10.2 and 10.3.

Year	Deregulatory event	Industry affected
1981	Decontrol of crude oil and refined petroleum products (executive order) Deregulation of radio (FCC)	Petrol and Natural Gas Entertainment
1982	Garn-St. Germain Depository Institutions Act AT&T settlement	Banking Communications
1984	Cable Television Deregulation Act Shipping Act	Entertainment Transportation
1987	Elimination of fairness doctrine (FCC)	Entertainment
1989	Natural Gas Wellhead Decontrol Act of 1989	Petrol and Natural Gas
1991	Federal Deposit Insurance Corporation Improvement Act	Banking
1992	Cable Television Consumer Protection and Competition Act Energy Policy Act FERC Order 636	Entertainment Petrol and Natural Gas Utilities
1993	Elimination of state regulation of cellular telephone rates Negotiated Rates Act	Communications Transportation
1994	Trucking Industry and Regulatory Reform Act Interstate Banking and Branching Efficiency Act	Transportation Banking
1995	Interstate Commerce Commission Termination Act	Transportation
1996	Telecommunications Act	Communications

Table 3, Panel B: Major deregulatory events and affected industries.

6.4 Figures 1 and 2: Capital liquidity and valuation variables

Read Figure 1:

- The vertical bars show total merger bids over time.
- Horizontal bars show the timing of industry merger waves.

- The line is the C&I loan-rate spread over the federal funds rate; lower values indicate higher capital liquidity.
- Industry waves cluster when capital liquidity is high.

Read Figure 2:

- The figure compares the rate-spread liquidity proxy with market-to-book and prior S&P 500 returns.
- The variables are related, which explains why high valuations can be correlated with waves even if liquidity is the deeper mechanism.

Economic message: These figures show why one must distinguish valuation from liquidity. Merger waves occur during boom-like periods, but the paper argues that the financing/liquidity component is central.

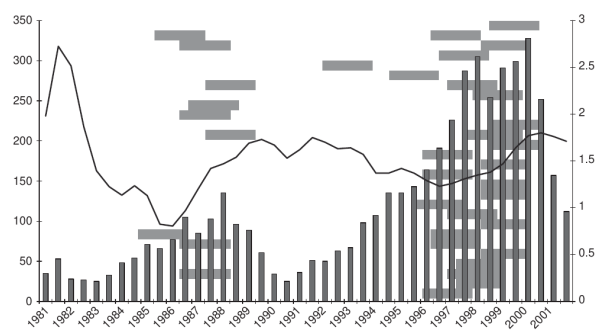


Fig. 1. Capital liquidity, industry merger waves and aggregate merger activity. The line is the spread between the average rate charged for commercial and industrial loans and the fed funds rate, reported in the Federal Reserve's Survey of Terms of Business Lending (right axis). This spread is measured in percentage points and proxies for low capital liquidity. The horizontal bars mark the timing of the industry-level merger wave periods in this study (e.g. the top-most bar represents an industry-level merger wave starting in the latter half of 1998 and ending in the first half of 2000). The vertical bars represent the

Figure 1: Capital liquidity, industry merger waves, and aggregate merger activity.

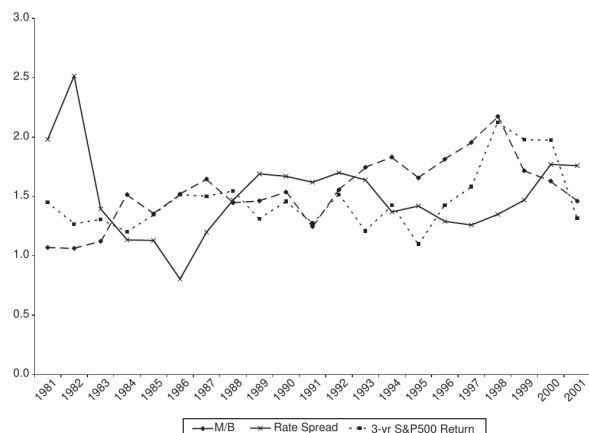


Fig. 2. Time-series relation between the rate spread variable and variables commonly used in behavioral explanations of merger waves. The rate spread is the spread between the average commercial and industrial loan rate and the Fed Funds rate, as collected by the Federal Reserve in its Survey of Terms of Business Lending. This spread is measured in percentage points and proxies for low capital liquidity. The M/B time series is the median market-to-book ratio of all firms on Compustat each year. Finally, 3-yr S&P500 Return is one plus the compounded prior three-year return on the S&P500 index. All series are

Figure 2: Rate spread, market-to-book, and S&P 500 return variables.

6.5 Table 4 and Figure 3: Predicting industry and aggregate merger waves

Read Table 4:

- Market-to-book alone predicts industry waves, but it loses explanatory power once the neoclassical variables are included.
- The C&I rate spread has a negative coefficient: tighter capital markets make waves less likely.
- Deregulatory events and the economic shock index are positive predictors.
- The interaction between the economic shock index and tight capital is negative: shocks are less likely to produce waves when financing is tight.
- Behavioral return variables add little explanatory power to the neoclassical model.

Read Figure 3:

- The figure shows that aggregate merger activity is strongly related to the share of bids coming from industries that are in merger waves.
- The paper reports a correlation of 0.85 between industry-wave involvement and aggregate merger activity.

Economic message: The formal prediction models support the shock-plus-liquidity view. Aggregate waves arise because industry waves bunch together when capital liquidity is high.

Table 4
Predicting merger waves
Logit models are used to predict when an industry will have a merger wave. The sample is 48 industries, each over 20 years (1981–2000). The dependent variable in the first four columns is equal to one if the industry-year is the beginning of a merger wave in that industry. The explanatory variables are measured at the end of year $t-1$. Market-to-book is the industry median market-to-book ratio, 3-year return and $\sigma(3\text{-year Return})$ are the median return in the industry for the three years ending at the end of year $t-1$ and the intra-industry standard deviation of that return, and the commercial and industrial (C&I) loan rate spread (spread above the fed funds rate) proxies for low capital liquidity. There is also a dummy variable selecting years that were preceded by a major deregulatory event. The economic shock index is the first principal component of the seven economic shock variables in the first column of Table 3. The shock index is also interacted with a dummy variable selecting years when market-to-book ratios are below their industry-specific time-series median or the C&I rate spread is above its time-series median (years of low capital liquidity).

The last three columns regress an indicator for aggregate merger activity on aggregate versions of the independent variables. There are 20 observations, one for each year from 1981–2000. The dependent variable identifies whether aggregate merger activity is in the top, middle, or bottom third of the time-series of merger activity over the sample period. Aggregate merger activity is defined as the fraction of firms in the population involved in merger activity (as a target or bidder) in a given year. Years in the top third can be thought of as merger wave years (the years are 1986–1988 and 1996–1999). The independent variables are weighted averages of the industry-level variables for that year, where the weights are the number of firms in the industry. For example, the economic shock index is the average economic shock index across all industries, weighted by the number of firms in each industry.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Intercept	-4.445 [<.0001]	-4.450 [<.0001]	-2.980 [<.0001]	-3.320 [<.0001]	-0.713 [0.074]	3.499 [0.001]	2.306 [0.173]
M/B _{$t-1$} ^a	0.847 [0.017]	0.840 [0.031]		0.165 [0.745]	2.172 [0.002]		0.520 [0.297]
3-year Return _{$t-1$}		0.152 [0.736]		-0.109 [0.850]	1.027 [0.216]		0.426 [0.463]
$\sigma(3\text{-year Return})_{t-1}$		-0.059 [0.846]		0.250 [0.492]	-0.790 [0.203]		0.015 [0.973]
C&I Rate Spread _{$t-1$}			-0.521 [0.032]	-0.567 [0.026]		-1.079 [0.004]	-0.998 [0.011]
Deregulatory Event _{$t-1$}			1.872 [0.025]	1.930 [0.023]		3.313 [0.291]	2.166 [0.488]
Econ Shock Index _{$t-1$}			0.452 [0.007]	0.369 [0.091]		1.137 [0.001]	1.000 [0.003]
Econ Shock Index _{$t-1$} * (Tight Capital)			-0.917 [0.001]	-0.881 [0.002]		-0.290 [0.083]	-0.205 [0.268]
Adj R ²					0.356	0.774	0.786
Pseudo-R ²	0.016	0.017	0.151	0.154			
Correlation of prediction with waves	0.079	0.075	0.240	0.248			

Table 4: Logit models predicting industry merger waves and aggregate activity.

6.6 Figure 4 and Table 5: Partial-firm acquisitions

Read Figure 4:

- Firm-level mergers and partial-firm acquisitions rise around industry merger waves.
- Cash partial-firm acquisitions also rise around waves.
- This pattern is consistent with broad asset reallocation, not merely stock-financed whole-firm merger timing.

Read Table 5:

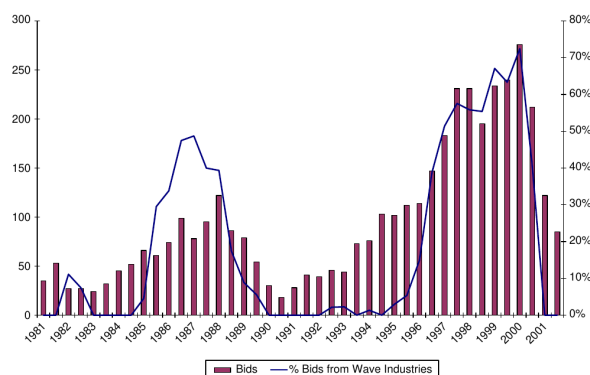


Fig. 3. Relation between industry merger waves and aggregate merger activity. The height of each bar represents the number of bids, shown on the left axis, with a deal value of at least \$50 million (2002 dollars) across all industries in that year. The line indicates the percent of all bids, shown on the right axis.

Figure 3: Industry merger waves and aggregate merger activity.

- Panel A shows strong correlations between merger activity and partial-firm acquisition activity.
- Panel B shows that merger bidders are more likely to be buyers in partial-firm acquisitions.
- Stock bidders are also more likely to make cash partial-firm acquisitions.

Economic message: This is one of the paper’s sharpest pieces of evidence. If firms are buying divisions for cash during waves, merger waves look like real asset-reallocation episodes rather than just overvalued-stock swap episodes.

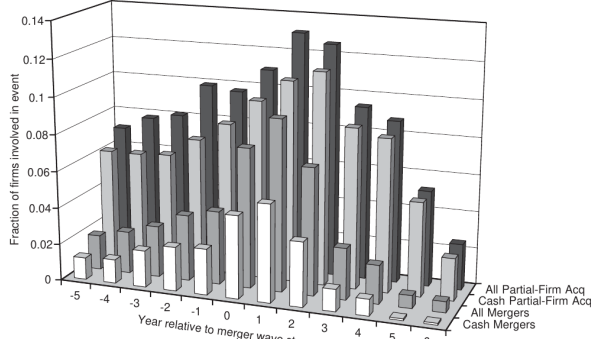


Figure 4: Merger and partial-firm acquisition activity around industry waves.

Table 5

(Panel A) The relation between merger activity and partial-firm acquisitions

The simple correlations between the fraction of firms in an industry that bid in a firm-level merger and are buyers in a partial-firm level acquisition over any 12-month period are presented. Partial-firm does not mean transactions for less than 100% of a selling firm’s outstanding equity. Rather, it refers to divisional transactions (outright purchases of an operational piece of the selling firm). The merger and partial-firm transactions are also split according to whether the method of payment was cash or stock. The correlations are calculated separately for all 48 industries and the cross-industry mean correlation is presented here along with its p -value, in brackets.

	Stock mergers	Cash mergers	Total partial-firm	Stock partial-firm	Cash partial-firm
Total mergers	0.728 [0.000]	0.902 [0.000]	0.369 [0.000]	0.179 [0.000]	0.341 [0.000]
Stock mergers		0.535 [0.000]	0.176 [0.000]	0.218 [0.000]	0.109 [0.019]
Cash mergers			0.331 [0.000]	0.124 [0.008]	0.273 [0.000]
Total partial-firm				0.312 [0.000]	0.890 [0.000]
Stock partial-firm					0.159 [0.000]

Table 5, Panel A: Correlations between merger activity and partial-firm acquisitions.

Table 5 (continued)

(Panel B) Logit models to predict which firms will be buyers in partial-firm acquisitions. The dependent variable takes a value of one in years in which the firm is a buyer in at least one partial-firm transaction, and is zero otherwise. In the first column, this variable is one for buyers in any partial-firm acquisition and in the final two columns it is one only for buyers in cash partial-firm acquisitions. The independent variables include three dummy variables indicating whether the firm was a bidder in a merger in that year, a cash bidder, or a stock bidder. The C&I rate spread for the year and a dummy variable set to one if the year is during the industry's merger wave are included in the specification. The remaining control variables are all measured in the prior year and are: cash to total assets, asset turnover, size (the log of sales), market-to-book of assets, cash flow scaled by beginning-of-period assets, leverage (total debt scaled by assets), and the one-year stock return.

	All partial acquisitions	Cash partial acquisitions	
Intercept	-2.811 [<.0001]	-3.429 [<.0001]	-4.299 [<.0001]
Bidder _{<i>t</i>}	0.863 [<.0001]	0.906 [<.0001]	
Cash bidder _{<i>t</i>}			0.874 [0.009]
Stock bidder _{<i>t</i>}			1.743 [<.0001]
C&I rate spread _{<i>t</i>}	-0.637 [<.0001]	-0.492 [<.0001]	-0.503 [<.0001]
In-wave dummy _{<i>t</i>}	9.128 [<.0001]	3.378 [<.0001]	3.378 [<.0001]
Cash _{<i>t-1</i>}	-0.196 [0.017]	-0.046 [0.651]	-0.036 [0.727]
Asset turnover _{<i>t-1</i>}	-0.257 [<.0001]	-0.425 [<.0001]	-0.427 [<.0001]
Size _{<i>t-1</i>}	0.336 [<.0001]	0.303 [<.0001]	0.307 [<.0001]
Market-to-book _{<i>t-1</i>}	0.017 [0.000]	0.015 [0.012]	0.015 [0.010]
Cash flows _{<i>t-1</i>}	0.117 [0.067]	0.360 [<.0001]	0.359 [<.0001]
Leverage _{<i>t-1</i>}	0.069 [0.174]	0.149 [0.010]	0.147 [0.011]
Stock return _{<i>t-1</i>}	0.079 [<.0001]	0.076 [<.0001]	0.076 [<.0001]
Pseudo- <i>R</i> ²	0.132	0.106	0.105
Firm-years	108183	108183	108183

Table 5, Panel B: Firm-level logit models for partial-firm acquisition buyers.

6.7 Table 6: Long-run stock-return performance

Read Panel A:

- The industry-level return tests do not show robust abnormal Fama–French adjusted returns before, during, or after waves.
- Unadjusted value-weighted returns show high returns before and during waves and lower returns after waves, but this pattern is not robust to equal weighting.

Read Panel B:

- Bidders in stock-financed mergers show some value-weighted underperformance.
- The underperformance is concentrated among stock bidders in waves in one specification.
- The evidence is not robust to equal weighting and is not significantly different from the performance of non-wave stock bidders.

Economic message: The return evidence is weak for the behavioral hypothesis. It does not show the robust post-wave reversal that one would expect if waves were primarily driven by market overvaluation.

Table 6

(Panel A) Calendar time-based regressions of long-run stock return performance for the 28 industries. The dependent variables in the VW and EW columns are the value-weighted and equally weighted returns, respectively, for month t of the portfolio of merged companies. Market, small minus big, and high minus low are the monthly factor realizations of the Fama and French (1993) factors. The estimation procedure is described in Section 4. Abnormal performance is detected in the intercept or dummy variable. The regressions are estimated separately for the time series of each of the 28 industries over the entire sample period. The median coefficient estimates from those 28 regressions and the p -values for a test of whether the median is different from zero are presented here. Pre-wave is a dummy variable identifying the 12 months immediately prior to a merger wave, In-wave is a dummy variable identifying the 24-month wave period, and post-wave is a dummy variable identifying the 36 months following the end of the wave.

	VW	EW	VW	EW
Intercept	0.139 [0.459]	-0.016 [0.845]	0.751 [0.001]	0.787 [0.001]
Pre-wave	0.155 [0.447]	0.069 [0.665]	0.895 [0.001]	-0.159 [0.627]
In-wave	-0.149 [0.998]	-0.134 [0.511]	1.195 [0.001]	0.415 [0.150]
Post-wave	0.185 [0.669]	0.351 [0.260]	-0.865 [0.072]	-0.610 [0.103]
Market	1.024 [0.001]	0.994 [0.001]		
Small minus big	0.052 [0.275]	0.892 [0.001]		
High minus low	-0.065 [0.301]	0.196 [0.001]		

Table 6, Panel A: Calendar-time long-run returns for the 28 wave industries.

Table 6 (continued)

(Panel B) Calendar time-based regressions of long-run stock return performance following a completed merger

The calendar time estimation is described in Section 4. Stock is a dummy variable taking the value of one for mergers in which stock was the method of payment and zero otherwise. In wave stock and non-wave stock dummies identify stock-payment mergers taking place inside or outside and industry merger wave. Market, small minus big, and high minus low are the monthly factor realizations of the Fama and French (1993) factors. The observations are monthly returns for time series of portfolios containing bidders in different categories. The number of observations varies depending on the number of different samples considered and the number of calendar months in the estimation. In columns 1 and 2, there are 237 total months in the time series of returns for the portfolio of cash bidders and 248 for the portfolio of stock bidders. In columns 3 and 4, there are 248 months of data for portfolios of non-wave stock bidders, 215 months of data for portfolios of wave stock bidders, and the remaining 237 observations are the time series of months for the cash bidders. Columns headed VW use value-weighted returns and those headed EW use equally weighted returns as the dependent variables. P -values from heteroskedasticity-consistent tests of the coefficients are presented.

	VW	EW	VW	EW
Intercept	0.254 [0.132]	-0.213 [0.280]	0.233 [0.166]	-0.223 [0.254]
Stock	-0.396 [0.057]	-0.179 [0.567]		
In-wave stock			-0.667 [0.021]	-0.458 [0.352]
Non-wave stock			-0.258 [0.230]	-0.074 [0.783]
Market	1.044 [<.001]	1.239 [<.001]	1.057 [<.001]	1.252 [<.001]
Small minus big	-0.179 [<.001]	0.341 [<.001]	-0.180 [<.001]	0.336 [<.001]
High minus low	-0.047 [0.323]	0.372 [<.001]	-0.004 [0.922]	0.382 [<.001]
Adj R^2	0.904	0.844	0.878	0.824
N	485	485	700	700

Table 6, Panel B: Calendar-time long-run returns for completed-merger bidders.

6.8 Table 7: Operating performance and analyst forecast changes

Read:

- Table 7 regresses post-merger performance on pre-merger performance and an in-wave dummy.
- Mergers in waves are not associated with worse post-merger operating performance.
- Sales growth is significantly higher for wave mergers in several specifications.
- Analyst long-term growth forecast revisions are more positive for wave mergers.
- Adding a contractionary-wave dummy does not overturn the conclusion.

Economic message: Mergers in waves do not look like low-synergy deals caused only by overvalued stock. They look at least comparable, and in some dimensions better, than mergers outside waves.

	Individual observations					
	Profitability	Asset turnover	ROA	Sales growth	M/B	LTG forecast
Intercept	-0.016 [0.073]	0.017 [0.094]	-0.015 [<.0001]	-0.003 [0.683]	-0.035 [0.289]	0.410 [0.067]
Pre-merger Measure	0.590 [<.0001]	0.608 [<.0001]	0.449 [<.0001]	0.159 [<.0001]	0.362 [<.0001]	0.986 [<.0001]
In wave	-0.010 [0.522]	0.024 [0.148]	0.004 [0.424]	0.024 [0.043]	0.069 [0.194]	0.372 [0.059]
Adj R^2	0.330	0.517	0.296	0.032	0.204	0.895
Obs	1323	1323	1323	1323	1323	613
Intercept	-0.016 [0.073]	0.017 [0.095]	-0.015 [<.0001]	-0.003 [0.691]	-0.035 [0.289]	0.433 [0.054]
Pre-merger Measure	0.590 [<.0001]	0.609 [<.0001]	0.449 [<.0001]	0.157 [<.0001]	0.362 [<.0001]	0.984 [<.0001]
In wave	-0.004 [0.813]	0.031 [0.081]	0.006 [0.294]	0.035 [0.007]	0.065 [0.266]	0.292 [0.169]
Contraction	-0.025 [0.383]	-0.033 [0.296]	-0.007 [0.428]	-0.048 [0.033]	0.020 [0.846]	0.346 [0.329]
Adj R^2	0.330	0.517	0.296	0.035	0.204	0.895
Obs	1323	1323	1323	1323	1323	613

Table 7, part 1: Individual-observation operating-performance and forecast regressions.

	Group means					
	Profitability	Asset turnover	ROA	Sales growth	M/B	LTG forecast
	-0.008 0.503	0.025 [0.106]	-0.014 [0.004]	0.004 [0.718]	-0.020 [0.686]	0.548 [0.114]
	0.970 [<.0001]	0.482 [<.0001]	0.243 [0.070]	0.029 [0.814]	0.319 [0.001]	0.975 [<.0001]
	-0.005 [0.804]	0.021 [0.396]	0.003 [0.657]	0.027 [0.092]	0.073 [0.272]	0.390 [0.084]
	0.479 56	0.450 56	0.028 56	0.022 56	0.220 56	0.972 55
	-0.008 [0.513]	0.025 [0.107]	-0.014 [0.003]	0.006 [0.618]	-0.016 [0.732]	0.631 [0.070]
	0.988 [<.0001]	0.487 [<.0001]	0.254 [0.048]	0.004 [0.976]	0.309 [0.001]	0.969 [<.0001]
	0.003 [0.880]	0.026 [0.324]	0.005 [0.558]	0.039 [0.021]	0.070 [0.330]	0.306 [0.198]
	-0.035 [0.351]	-0.023 [0.616]	-0.006 [0.675]	-0.052 [0.071]	0.019 [0.876]	0.396 [0.320]
	0.489 59	0.438 59	0.022 59	0.062 59	0.207 59	0.969 57

Table 7, part 2: Group-mean operating-performance and forecast regressions.

7 Short summary

- The paper asks whether merger waves are caused by market timing or by real industry shocks.
- It identifies 35 industry-level merger waves from 1981 to 2000 using 48 Fama–French industries and a simulation-based wave definition.
- Economic shocks, deregulation, and technological changes frequently precede industry waves.
- Shocks are much more likely to create merger waves when capital liquidity is high.
- The key liquidity proxy is the spread between commercial and industrial loan rates and the federal funds rate.
- Market-to-book ratios are high before waves, but their predictive power largely disappears after including real shocks and capital liquidity.
- Aggregate merger waves are mostly the clustering of industry-level waves.
- Cash partial-firm acquisitions rise during waves, which supports a real asset-reallocation explanation.
- Return evidence does not show robust post-wave underperformance.
- Operating performance and analyst forecast revisions do not suggest that wave mergers are worse than non-wave mergers.
- The main conclusion is that merger waves are driven by industry shocks plus capital liquidity, not primarily by attempts to time market misvaluation.

8 Conclusion

8.1 Core conclusion

Harford argues that merger waves are best explained by a neoclassical reallocation model with a capital-liquidity condition. Economic, technological, and regulatory shocks create reasons for industries to restructure. When financing is easy and capital liquidity is high, these shocks produce concentrated merger waves.

8.2 Economic intuition

Mergers are a way to move assets to higher-value uses. But moving assets is costly. When credit is tight or buyers are financially constrained, even a large industry shock may not generate a wave. When capital is liquid, the same type of shock can lead to rapid asset reallocation through mergers and partial-firm acquisitions.

8.3 Why this differs from behavioral market timing

The behavioral explanation emphasizes overvalued bidders using equity to buy real assets. Harford finds that valuation variables alone are not enough. Once liquidity and industry shock variables are included, market-timing variables add little. In addition, cash divisional acquisitions rise in waves, which is not a natural prediction of the overvalued-stock mechanism.

8.4 Implications

The findings imply that aggregate merger waves should be understood as episodes of real restructuring enabled by favorable financing conditions. High market valuations may still matter, but mainly because they coincide with high liquidity and lower transaction costs.

8.5 Limitations

The evidence is observational. Market timing may still explain some individual transactions, and market-to-book remains difficult to interpret because it can reflect both misvaluation and real growth opportunities. The paper's stronger claim is that market timing is not the main driver of merger waves.

8.6 Takeaway

The enduring takeaway is that merger waves require both motive and means: industry shocks provide the motive for asset reallocation, and capital liquidity provides the means to execute that reallocation at wave scale.